# CineMatch

Hybrid Movie Recommender System

**PROJECT EXECUTION PLAN**

5 Members | 8 Weeks | Python + Streamlit

| Project | Team | Duration | Methodology |
|---|---|---|---|
| CineMatch | 5 Members | 8 Weeks | Collaborative Development |

## 1 Project Team & Task Distribution

The project will be developed by five team members. Each member is responsible for a specific part of the project, while all members collaborate during integration, testing, and final delivery.

### 1 Mohamed Salah Abozaid

Data Engineering & Analysis

**RESPONSIBILITIES**

- Data Collection and Integration.
- Data Cleaning and Preprocessing.
- Exploratory Data Analysis (EDA).
- Preparing clean datasets for model development.

**MAIN DELIVERABLES**

- Cleaned movie dataset.
- Integrated ratings and movie metadata.
- Data analysis notebook and visualizations.

### 2 Ahmed Reda Fathy Salim

Content-Based Filtering & NLP

**RESPONSIBILITIES**

- Content-Based Filtering.
- Text preprocessing and feature engineering.
- TF-IDF Vectorization.
- Cosine Similarity calculations.
- VADER Sentiment Analysis.

**MAIN DELIVERABLES**

- Content-Based Recommendation Model.
- Movie similarity matrix.
- Content-Based Filtering notebook.

### 3 Mohamed Mahmmud Ibrahim Hamed Sakr

Collaborative Filtering

**RESPONSIBILITIES**

- Collaborative Filtering.
- User-Item Interaction Matrix.
- User similarity calculations.
- Rating prediction and recommendation logic.

**MAIN DELIVERABLES**

- Collaborative Filtering Model.
- User similarity matrix.
- Collaborative Filtering notebook.

### 4 Abdelrhman Abdelsamad Elessawy Abdelsamad

Hybrid System & Evaluation

**RESPONSIBILITIES**

- Hybrid Recommendation System.
- Combining Content-Based and Collaborative Filtering.
- Recommendation score calculation.
- Model evaluation and performance testing.

**MAIN DELIVERABLES**

- Hybrid Recommendation Engine.
- Top-N recommendation results.
- Model evaluation report.

### 5 Mohamed Sayed Mahmoud Mohamed
Streamlit Application & UI/UX

**RESPONSIBILITIES**
- Streamlit Web Application.
- User Interface and User Experience (UI/UX).
- Movie search and rating functionality.
- Displaying recommended movies and movie information.
- Application integration and deployment.

**MAIN DELIVERABLES**
- Interactive Streamlit Application.
- Application interface and functionality.
- Deployment and final application demo.

### Shared Responsibilities
- Reviewing and integrating each other's work.
- Testing the complete recommendation system.
- Fixing integration issues and bugs.
- Preparing the final presentation.
- Documenting the project and explaining its implementation.

## 2 Project Timeline

The project is planned to be completed over 8 weeks.

| Week | Main Activities | Responsible Members |
|---|---|---|
| **Week 1** | Data Collection and Integration | Mohamed Salah |
| **Week 2** | Data Cleaning and EDA | Mohamed Salah |
| **Week 3** | Content-Based Filtering and NLP | Ahmed Reda |
| **Week 4** | Collaborative Filtering | Mohamed Sakr |
| **Week 5** | Hybrid Model Development | Abdelrhman |
| **Week 6** | Model Integration and Evaluation | All Model Developers |
| **Week 7** | Streamlit Application Development | Mohamed Sayed |
| **Week 8** | Final Testing, Deployment and Presentation | All Members |

*Note: Members should communicate regularly and share intermediate results to avoid delays during integration.*

## 3 Testing & Validation

The system will be tested using the following scenarios to ensure that recommendations work correctly and the application handles different user situations.

| Test Case | Scenario | Expected Result |
|---|---|---|
| **TC-01** | A new user has not rated any movies. | The application asks the user to rate at least 3 movies and displays a friendly message. |
| **TC-02** | A user rates 3 movies from different genres. | The system generates a personalized Top-10 recommendation list. |
| **TC-03** | A selected movie has insufficient collaborative data. | The Content-Based engine provides alternative recommendations. |
| **TC-04** | A user updates movie ratings multiple times. | The application updates recommendations without unnecessarily reloading large model files. |

### Performance Evaluation

- Recommendation relevance.
- Accuracy of predicted ratings where applicable.
- Precision@10 for recommendation ranking.
- Response time.
- Memory usage.
- Application stability.

## 4 Technology Stack

| Category | Technologies |
|---|---|
| **Programming Language** | Python |
| **Data Processing** | Pandas, NumPy |
| **Data Visualization** | Matplotlib, Seaborn |
| **Machine Learning** | Scikit-learn |
| **NLP** | TF-IDF, NLTK (VADER) |
| **Similarity Calculations** | Cosine Similarity |
| **Sparse Matrix Operations** | SciPy |
| **Web Application** | Streamlit |
| **Data Sources** | MovieLens, TMDB, IMDb |
| **Version Control** | Git, GitHub |
| **Deployment** | Streamlit Community Cloud |

## 5 Project Deliverables

The final project submission will include:

1. Cleaned and integrated datasets.
2. Data Collection and Cleaning notebook.
3. Exploratory Data Analysis notebook.
4. Content-Based Filtering notebook.
5. Collaborative Filtering notebook.
6. Hybrid Recommendation System notebook.
7. Trained and serialized recommendation components.
8. Streamlit Web Application.
9. Project documentation and README.
10. Final presentation and live demonstration.

*All deliverables will be reviewed and verified before being marked as completed.*

## 6 Future Improvements

The following features may be considered for future versions of CineMatch:

**Advanced Recommendation Models**
Implement SVD or SVD++ to improve collaborative recommendations.

**Improved User Experience**
Add more filtering options, movie details, trailers, and personalized categories.

**Live TMDB Integration**
Retrieve updated movie posters, information, and trailers through the TMDB API.

**Deployment Improvements**
Use Docker and GitHub Actions for easier deployment and automated testing.

*These improvements are optional and will not affect the main project delivery.*

## 7 Final Project Goal

FINAL PROJECT GOAL

**The main goal is to deliver a fully functional Hybrid Movie Recommendation System that combines Content-Based Filtering and Collaborative Filtering into one application.**

Each team member will contribute to a specific part of the development process, ensuring that all components work together to provide users with personalized and relevant movie recommendations.